

Helpdesk Hardware Troubleshooting

Assignment Answers

1. Laptop will not power on even when plugged in

Step 1 — Check the wall outlet: Test the outlet with a known-working device or try another outlet. This checks whether mains power is available.

Step 2 — Inspect the charger and cable: Look for cuts, bent connectors, fraying, unusual heat, or burn marks. Check that the adapter is firmly connected; do not use a damaged charger.

Step 3 — Check the adapter indicator: If the charger has an LED, see whether it lights when plugged into the wall and whether it changes when connected to the laptop. A missing or changing light can suggest adapter or laptop-side trouble.

Step 4 — Disconnect peripherals and perform a power reset: Unplug USB devices and the charger. If the manufacturer permits, hold the power button for 15–20 seconds, reconnect the charger, and try again. This can clear a temporary power-state issue.

Step 5 — Check charging port, battery, and known-good adapter: With power disconnected, inspect the charging port for looseness or damage. If available, try a compatible, correctly rated known-good charger. If it still shows no lights or response, the battery, DC-in jack, power circuitry, or motherboard may need technician testing.

2. Monitor blank, but the PC appears to be running

1. Confirm the monitor is powered on and its power indicator is lit. Check brightness and select the correct input source (HDMI, DisplayPort, etc.).
2. Reseat the video cable at both the monitor and PC. Check for bent/damaged connectors; try a known-good cable and another monitor or display if available.
3. Make sure the cable is connected to the correct output. If the PC has a discrete graphics card, connect the monitor to the graphics card's ports rather than the motherboard ports.
4. Restart the PC and observe startup indicators: manufacturer logo, POST beeps, diagnostic LEDs, or error messages. If there is no logo at all, the issue may be before Windows starts.
5. If comfortable and the PC is powered off and unplugged, check that the RAM and graphics card are firmly seated and that GPU power connectors are attached. Do not open the power supply.
6. If the picture appears but goes blank after Windows loads, try the Windows display reset shortcut Win + Ctrl + Shift + B, then check display mode with Win + P and choose the appropriate display option.
7. If still blank, test with integrated graphics (only if supported) or a known-good graphics card, one change at a time. Record any beep/LED codes and escalate for hardware testing if needed.

3. Wireless mouse unresponsive while keyboard works

1. **Check mouse power:** Turn the mouse switch on, replace/recharge the battery, and check its power indicator if present.
2. **Check the wireless receiver or pairing:** For a USB dongle mouse, unplug and reconnect the receiver; try a direct USB port rather than a hub. For Bluetooth, confirm Bluetooth is enabled and the mouse is paired.
3. **Test another USB port:** Move the receiver to another port. If it works there, the original port or hub may be faulty; test the original port with another known-working USB device.
4. **Test the mouse on another computer:** If the mouse fails on a second system too, the mouse, battery, or receiver is likely at fault. If it works elsewhere, focus on the original computer's settings or drivers.
5. **Check system settings and drivers:** Use the touchpad or keyboard to check Bluetooth/device settings, Device Manager, and any mouse-disabled or accessibility settings. Restart the computer and update/reinstall the relevant device driver if needed.

Isolation summary: Mouse fails on multiple computers = likely mouse/receiver/battery; receiver fails only in one port = likely port/hub; mouse works on another PC but not this one = likely pairing, driver, or system configuration.

4. Gaming café PC overheats and shuts down: three hardware causes

Possible cause	How to check
1. Dust-clogged heatsink or air vents	Power down and unplug the PC. Inspect intake vents, filters, CPU/GPU heatsinks, and fans for dust buildup. Clean with appropriate compressed air while preventing fans from spinning freely; ensure airflow is not blocked.
2. Failed or obstructed cooling fan	Check whether CPU, case, and GPU fans spin when expected and listen for grinding or rattling. Verify the CPU fan is connected to the correct motherboard header. Replace a failed fan with a compatible part.
3. Poor heatsink contact or dried thermal paste	Check CPU/GPU temperature trends using trusted monitoring tools if available. If temperatures rise rapidly, a loose cooler or degraded thermal interface may be responsible. A trained technician should reseal the cooler and apply the correct amount of suitable thermal paste according to the manufacturer's instructions.

Safety note: Shut down and disconnect power before opening a computer or reseating components. Never open a power supply. If there is a burning smell, smoke, or unusually hot components, stop using the system and refer it to a qualified technician.